

1.0 Introduction / Background

The purpose of this report is majorly two-fold. To investigate the difference in job satisfaction between workers who use a computer at work and those who do not; and examine how workers' views of working in an organisation and how good they perceive managers influence their job satisfaction. This analysis is important because employees are viewed as the most important resource in an organisation. And their satisfaction is positively correlated with firm/ organisational performance (Meier and Spector (2015; Sainju, Hartwell and Edwards, 2021). For example, unsatisfied employees can increase turnover rates (Pablo, 2023), which can significantly impact organisations' operations and finances (Kim et al., 2023).

➤ Data

Data used for this analysis was based on the Work and Employment Relations (WERS) 2004 survey (abridged version). The dataset contains 126 variables (columns) and 11, 226 cases (rows), with each row containing a response from an employee. These responses were obtained through a cross-sectional survey.

➤ Measures

The dependent (outcome) and independent (explanatory) variables were already given for this assessment. While there are a lot of variables in the dataset that could be controlled for this analysis, GENDER (SEX) and AGE were chosen as the control variables. The rationale is that the literature has broadly suggested that gender (Andrade, Miller and Westover, 2022) and age (Cavanagh, Kraiger and Henry, 2020; Kollmann et al., 2020) affect workers' job satisfaction.

- Dependent Variables - Job Satisfaction (a8g – a8g)
- Independent Variables - How good managers are (b8a – b8c) + Your views on working here (c1a-c1c)
- Control Variables: Gender (e1) + Age (e2)

2.0 Methodology

➤ Data Exploration

Before analysis, the relevant variables (16) in the original dataset were first extracted (White, Royston and Wood, 2011) to explore the degree- and pattern of missingness. That is if the data are missing completely at random, missing at random (MAR) and missing not at random (MNAR).

➤ Data Cleaning (Multiple Imputation)

To replace the missing data in the variables, the multiple imputation technique (regression method) was used. This approach was taken because it is an effective technique for dealing with missing data, especially data missing not at random (MNAR). It works by running iterations and replacing missing data with data likely similar to the missing data. In the context of this analysis, we assumed it to be better than other approaches like listwise/pairwise deletion because these methods are not suitable for dealing with MNAR situations. Furthermore, single imputation like expectation maximization (EM) is also not suitable in this context as it is useful only in cases of MCAR. The main disadvantage of these alternative methods is the issue of bias which MI can address. Moreover, our main focus was on the pattern of missingness rather than the degree of missingness. For example, Tabachnick and Fidell (2012) argue that the impact of the mechanism of missingness on research results far outweighs the size of missingness.

For the MI process, we performed 20 imputations at 10 iterations because of the percentage of missing cases in the extracted dataset (White, Royston and Wood, 2011). Additionally, at lower imputation, enough values were not generated to capture/replace all the missing values. Hence, we were only able to generate completeness at 20 imputations.

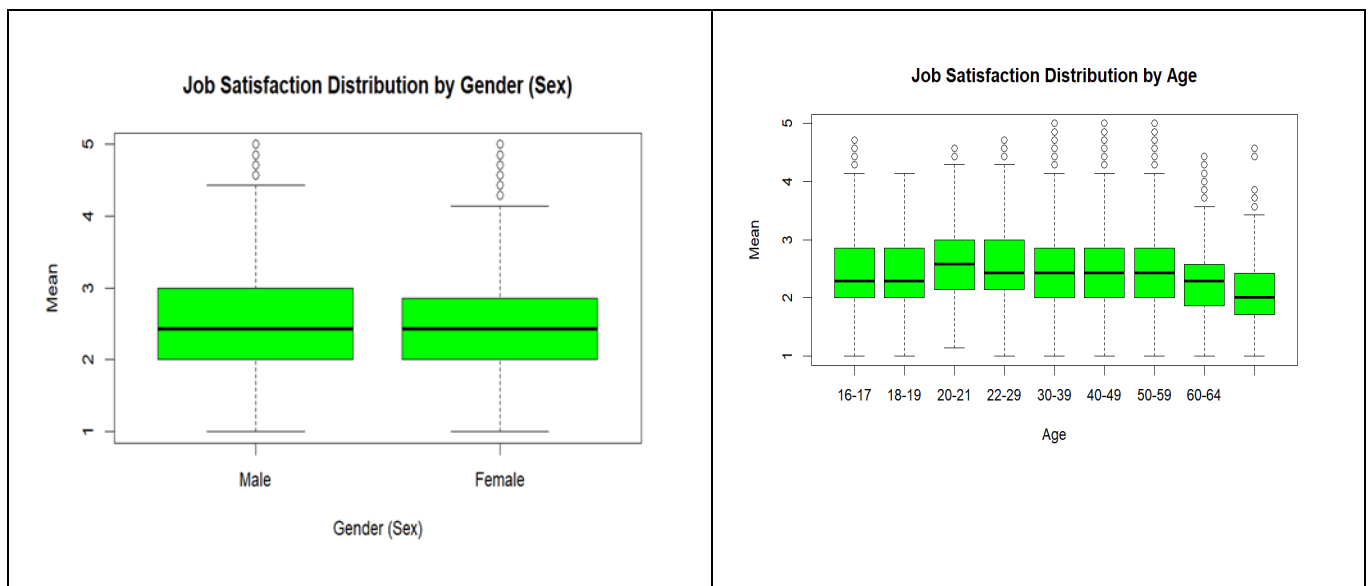
➤ Analysis

All analyses were conducted using R programming statistical software. The codes were written in R Markdown and have been attached to this document.

3.0 Result

➤ Descriptive Statistics

Figure 3.1a & b: Pooled Mean Job Satisfaction by Gender and Age



➤ Independent Sample t-test

Figure 3.2: Pooled Mean Job Satisfaction by Computer Use at Work

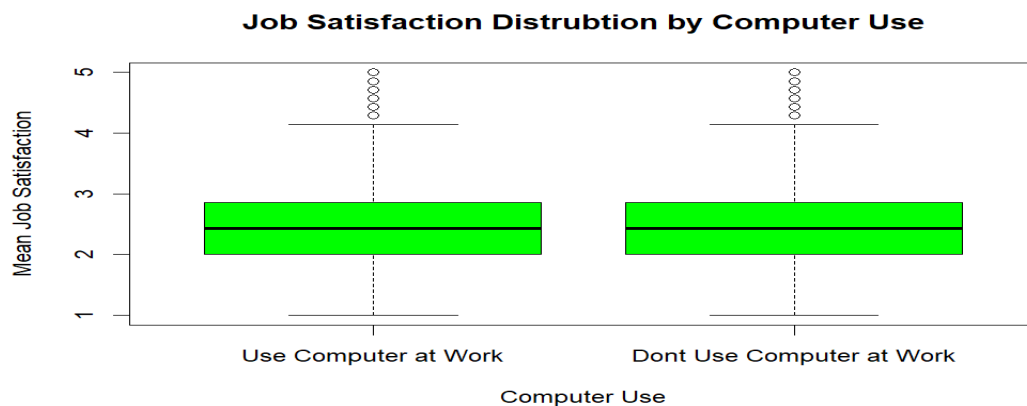


Table 3.1: Levene's Test for Homogeneity of Variance (centre = median)

	Df	F value	Pr(>F)
	1	33.936	5.702e-09 ***
Group	224518		

Signif. codes: 0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Null Assumption: Variances across all samples are equal.

Table 3.2: Welch Two Sample t-test

Df	t-value	p-value	sample estimates		95 percent confidence interval	
			mean in group 0	mean in group 1	lower	upper
96695	4.0031	6.26E-05	2.487697	2.474191	0.006893159	0.020118332

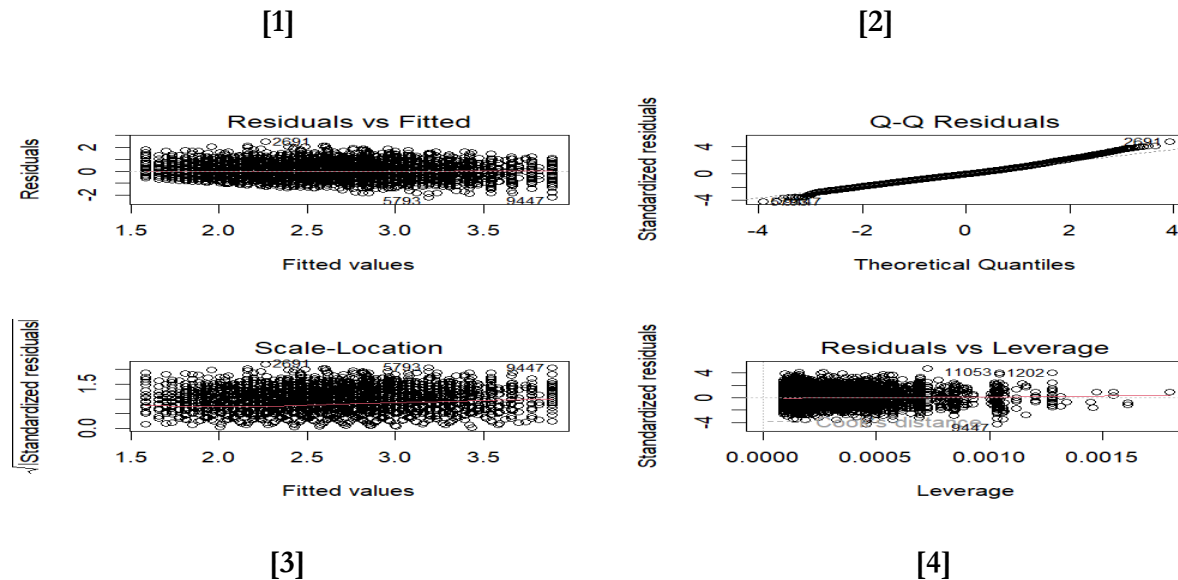
Alternative hypothesis: True difference in means between group 0 and group 1 is not equal to 0

Table 3.3: Regression Results

Regression Models					
		Dependent Variables			
		Job Satisfaction			
	extracted_data.reg	reg.model.mi	reg.model.mi	reg.model.mi3	reg.model.mi4
		1	2	3	4
Intercept	0.97***(0.02)	1.02***(0.01)	1.03*** (0.02)	1.12*** (0.03)	1.13*** (0.03)
mean_Manview	0.23***(0.01)	0.22***(0.01)	0.22*** (0.01)	0.23*** (0.01)	0.23*** (0.01)
mean_Workersview	0.35***(0.01)	0.35***(0.01)	0.34*** (0.01)	0.34*** (0.01)	0.34*** (0.01)
Control					
Gender			-0.01(0.01)		-0.01 (0.01)
Age				-0.02*** (0.00)	-0.02*** (0.00)
Adj. r-square	0.47	0.43	0.43	0.43	0.43

Level of significance: * = $p < .05$; ** = $p < .01$; *** = $p < .001$

Table 3.3: Assumptions for Linearity, Normality and Heteroskedasticity



4.0 Alternative Approaches to Examining the relationships between job satisfaction, views of working at an organisation and perception of how good managers are.

➤ Regression Approach

Regression can also be used to investigate the relationship between the outcome and explanatory variables as contained in this analysis. Regression analysis involves building models for explanatory and predictive purposes.

To conduct regression, it is important to first analyse the correlation between the variables to know if there is a relationship. While this does not entirely determine if regression should or can be conducted, it helps in shaping the analysis going forward. In the context of this study, job satisfaction (outcome variable) will be regressed against the two explanatory variables (how good managers are and workers' view of working in the organisation), after which the coefficient estimates and adjusted r-square will be recorded. Following this, more models are going to be built factoring in the effect of control variables (especially Age and Gender). How the result will be interpreted will be based on the nature of the data cleaning technique. For example, if the multiple imputation technique (MI) is used, the number of regressions generated will be equal to the imputation number, and the pooled average of the entire regressions becomes the focus of the model and not the individual regressions that make up the model. After the models are generated, the best-fit model will be selected based on the extent to which they can explain the variation of the outcome variable. Finally, the model will be subjected to assumption testing to test their reliability.

➤ Exploratory Factor Analysis (EFA)/Structural Equation Modelling (SEM)

Structural equation modelling (SEM) framework is an approach or technique that can be used to investigate the relationship between variables and their covariances. The use of SEM is not new in the academic literature. One key benefit of SEM is that it can be used to explore a variety of models to examine fit. With SEM, the interrelationship between variables can be explored either independently or with the addition of mediation (Mensah, 2019) and control.

In the context of this analysis, SEM can be used by first conducting an exploratory factor analysis (EFA) on the relevant variables using SPSS to examine factor loadings, communalities and eigenvalue with a view to identifying the optimal number of factors for retention in the construct. This step is important because it helps validate the factors before SEM analysis. Following this, an SEM model will be developed in SPSS Amos to examine the relationship between the retained outcome and explanatory variables in the construct, while also specifying the effect of the control variables.

5.0 Discussion

➤ Preliminary Exploration

Preliminary exploration of the key variables (See **Appendix A**) shows that all the key variables in the dataset have missing values with over 19% of the cases/observations missing at least one value (meaning only around 80% of the cases have complete data). From the distribution of missingness, the percentages of missing values range from 0.3 to 10.8%. For example, b8c (allowing employees or employee representatives to influence final decisions) shows the highest number of missing values at 10.8% with e1 (age) and gender (e1) the lowest with 0.3 and 0.4 respectively.

A look at the pattern of missingness in **Appendix C** suggests that the missing value tends to show some monotonic relationship between patterns and the variables indicating that the values are probably not missing at random. This was further validated by the MCAR little test (See **Appendix B**). The test suggests that the missing data are not missing completely at random because the p-value (0.00) is statistically significant thereby inconsistent with MCAR assumptions (fail to accept the null hypothesis).

After imputation, all missing values were replaced (See **Appendix D**).

➤ Descriptive Statistics

Figures 3.1a & b show the pooled mean job satisfaction of workers by gender and age respectively. The boxplot (1a) suggests that men (2.53) have a higher mean job satisfaction than female (2.44). Additionally, the boxplot (1b) shows fluctuations in mean job satisfaction across all age groups with workers with ages 65 and more the lowest and 20-21 the highest. (See details in Appendix E and F). The overall mean job satisfaction for all workers is 2.48.

➤ Independent Samples t-test

The result of the descriptive statistics shows that the group (0) which uses computers at work has a slightly higher job satisfaction value (2.4867) than those who don't use computers (group 1 - 2.4658) (See **figure 3.2**).

To investigate the difference in job satisfaction between those who use computers and those who don't, an independent samples t-test was conducted. An independent samples t-test is a statistical technique that is utilised in comparing the means of two independent groups to know if there is any significant difference.

Hypothesis: The Null hypothesis (H0) - There is no difference in job satisfaction between those who use computers and those who do not.

Alternative hypothesis (H1) - There is a difference in job satisfaction between those who use computers and those who do not.

Before conducting the test, the conditions for independent samples t-test were examined. These are, [1] the two groups are independent; [2] the variables are scaled in intervals; [3] the variables are normally distributed; and [4] the variance within the group is similar. Lavene's test was first conducted to test the homogeneity assumption (equality of variance) of the two groups. The result from the test suggests that there is no variance of equality between the 2 groups as the p-value is statistically significant (5.7×10^{-9}) which is essentially 0). Therefore, we can accept the null hypothesis of equal variance as the assumption of homogeneity of variance has been violated (See **Table 3.1**).

As a result of the violation of homogeneity of variance, Welch's test was carried out instead of the conventional t-test to test the above hypothesis. This is because the Welch test does not assume equality of variance. The p-value (6.255×10^{-5}) from Welch's test shows statistical significance, hence, suggesting that the groups differ with respect to their means. That is, those who use computers at work (group 0 - 2.49) and those who do not (group 1 - 2.47). As a result, we can reject the null hypothesis (See **Table 3.2**).

➤ Regression

The result of the correlation coefficient between job satisfaction/how good managers and job satisfaction/workers' view of working here are 0.5704325 and 0.6053835 respectively, suggesting a somewhat moderate - strong relationship between the variables.

We built five (5) regression models – one (1) from the extracted data before imputations and four (4) from the imputed data. Each imputed regression model contains a set of twenty (20) regressions. The models represented in **Table 3.3** are the pooled average of these regressions.

All the explanatory variables in the regression models show statistical significance. However, the control variable (gender) did not show any show statistical significance in two (2) of the imputed models (**reg.model.mi2 and reg.model.mi4**). Additionally, another control variable (Age), shows a negative relationship with the outcome variable (job satisfaction). Meaning that an increase in age will be associated with a decrease in mean job satisfaction if other variables remain constant.

The coefficients of regression were consistent across all five models irrespective of controls. This suggests that for every one-unit increase in workers' view of how good managers are and their view

of the workplace, the pooled job satisfaction mean is expected to increase by 0.22 (0.23) to 0.34 (0.35) respectively (See **Table 3.3**).

We also observed a higher adjusted r-square in the model generated from the data before imputation (0.47) compared to the other four (4) imputed models. Suggesting a reduction in the explanatory power of the model post-imputation (0.43). This figure was consistent in all the imputed models suggesting that only 43% of the variation in job satisfaction can be explained by the explanatory variables. This reduction could be a result of the variation introduced to the dataset during imputations. We can argue that the model generated before the imputation overestimated the effect of the explanatory variables (workers' views of working in an organisation and how good they perceive managers) on the outcome variables (job satisfaction).

Since all our fitted imputed models have the same explanatory effect on job satisfaction (adj. r-square), we can conclude that Model 1 (reg.model.mi) is more appropriate. This is because the addition of control variables in other models did not have any effect on the explanatory power of the models.

Furthermore, following the robustness test, we tested for linearity, normality, multicollinearity and heteroskedasticity. From **Figure 3.4**, we can assume linearity between the outcome and explanatory variables [1] and normality [2]. The result from the Variance Inflation Factor (VIF) shows no multicollinearity problem as the values in the 20 imputed models are less than 2 which is generally accepted as good. However, the heteroskedasticity [3] suggests a violation of the assumption of constant error variance across all levels of the predicted values. While we believe that this does not necessarily invalidate the result, it can affect the coefficient estimates and standard errors. Hence, we acknowledge the limitations of the model, especially in making predictive analysis.

Note: It is important to acknowledge that the VIF could not be computed from the pooled model (reg.model.mi) in R, and thus the average of the 20 imputations models was done manually.

6.0 Conclusion

The finding of our study shows that there is a significant difference in job satisfaction between workers who use computers at work and those who do not. It also shows that there is a positive relationship between job satisfaction and workers' views of working in an organisation and how good they perceive managers, albeit moderate. Furthermore, the explanatory effect of these variables together on job satisfaction is low with both the pre-processed and imputed data, although slightly higher in the former.

7.0 Limitations

The models did not do a very good job of explaining job satisfaction. We believe job satisfaction is a very broad concept prone to endogeneity problems which can be influenced by lots of both observed and unobserved variables not used in this analysis. For example, research has shown that unobserved variables like ambition and resilience can influence job satisfaction. Thus, increasing the number of explanatory variables might potentially increase the explanatory power of the model. Additionally, the

Akaike Information Criterion (AIC) cannot be computed from pooled multiple imputed data. Hence, challenging for it to be used for predictive purposes.

Appendices

Appendix A: Summary Statistics of all Relevant Variables (Before Imputation)

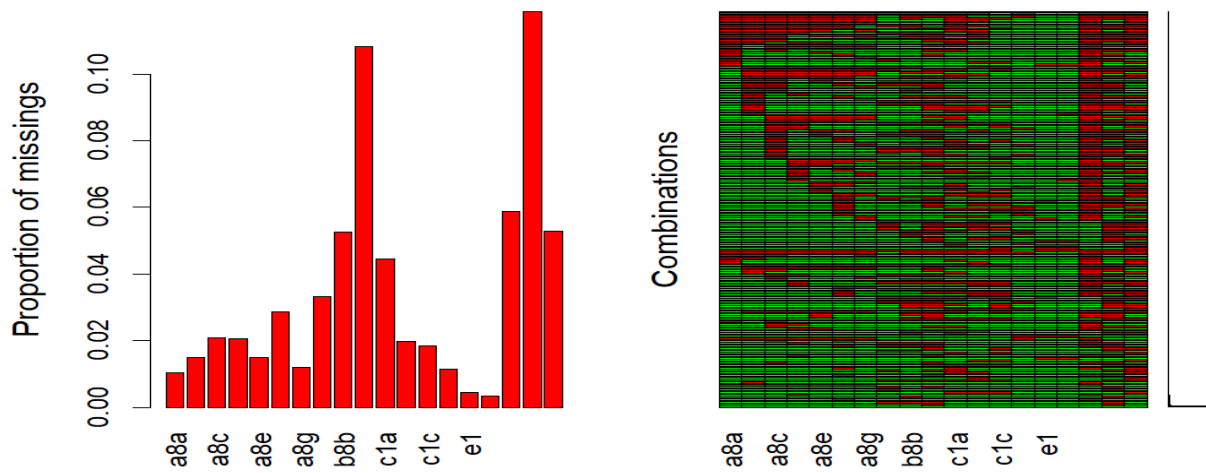
Variable s	Min	Max	Mean	Missing	
				Count	Percent
a8a	1	5	2.33	117	1
a8b	1	5	2.19	167	1.5
a8c	1	5	2.47	233	2.1
a8d	1	5	2.69	232	2.1
a8e	1	5	3.13	168	1.5
a8f	1	5	2.44	323	2.9
a8g	1	5	2.22	134	1.2
b8a	1	5	2.79	372	3.3
b8b	1	5	2.87	589	5.2
b8c	1	5	3.13	1216	10.8
c1a	1	5	2.47	501	4.5
c1b	1	5	2.11	221	2
c1c	1	5	2.34	208	1.9
e1	1	2		50	0.4
e2	1	9	5.55	37	0.3
e9_13	0	1		130	1.2

Note: e1 and e9_13 are categorical variables

Appendix B: MCAR little test

statistics	df	p.value	Missing.patterns
	<dbl>	<dbl>	<int>
4556.638	3203	0	280

Appendix C: Patterns of Missing Values



Appendix D: Summary Statistics of all Relevant Variables (After Imputation)

Variable s	Min	Max	Mean
.imp	1	20	10.5
.id	1	11226	
a8a	1	5	2.23
a8b	1	5	2.19
a8c	1	5	2.47
a8d	1	5	2.69
a8e	1	5	3.13
a8f	1	5	2.44
a8g	1	5	2.22
b8a	1	5	2.79
b8b	1	5	2.87
b8c	1	5	3.13
c1a	1	5	2.47
c1b	1	5	2.21
c1c	1	5	2.35
e1	1	2	
e2	1	9	5.55
e9_13	0	1	

Note: e1 and e9_13 are categorical variables

Appendix E: Job Satisfaction by Gender (Sex)

	Male	Female	Overall Mean
Job Satisfaction (Mean)	2.53	2.44	2.67

Appendix F: Job Satisfaction by Age

Code	Age	Job Satisfaction (Mean)
1	16-17	2.42
2	18-19	2.45
3	20-21	2.57
4	22-29	2.55
5	30-39	2.49
6	40-49	2.5
7	50-59	2.45
8	60-64	2.29
9	65 and more	2.11

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